

Experiment No._03

Title: Sensor and actuator interfacing with ESP32

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Batch: A2**Roll No.: 16010422078****Experiment No.:03****Aim:** and actuator interfacing with ESP 32

Resources needed: Internet, Raspberry Pi module, Sensors and Actuators

Theory:**Pre Lab/ Prior Concepts:**

The Internet of Things (IoT) is the network of physical devices, vehicles, home appliances and other items embedded with electronics, software, sensors, actuators, and connectivity which enable these objects to connect and exchange data. Each thing is uniquely identifiable through its embedded computing system but is able to inter-operate within the existing Internet infrastructure.

Role of Sensor in IoT:

Sensors are now found in a wide variety of applications, such as smart mobile devices, automotive systems, industrial control, healthcare, oil exploration and climate monitoring. Sensors are used almost everywhere, and now sensor technology is beginning to closely mimic the ultimate sensing machine the human being. The technology that allows this to happen is sensor fusion, which leverages a microcontroller to fuse the individual data collected from multiple sensors to get a more accurate and reliable view of the data than one would get by using the data from each discrete sensor on its own. Sensor fusion creates a situation in which the whole is much greater than the sum of its parts.

Role of Actuator in IoT:

An actuator is a component of a machine that is responsible for moving and controlling a mechanism or system, for example by opening a valve. In simple terms, it is a "mover".

An actuator requires a control device (controlled by control signal) and a source of energy. The control signal is relatively low energy and may be electric voltage or current, pneumatic, or hydraulic fluid pressure, or even human power. Its main energy source may be an electric current, hydraulic pressure, or pneumatic pressure. The control device is usually a valve. When it receives a control signal, an actuator responds by converting the source's energy into mechanical motion. In the electric, hydraulic, and pneumatic sense, it is a form of automation or automatic control.

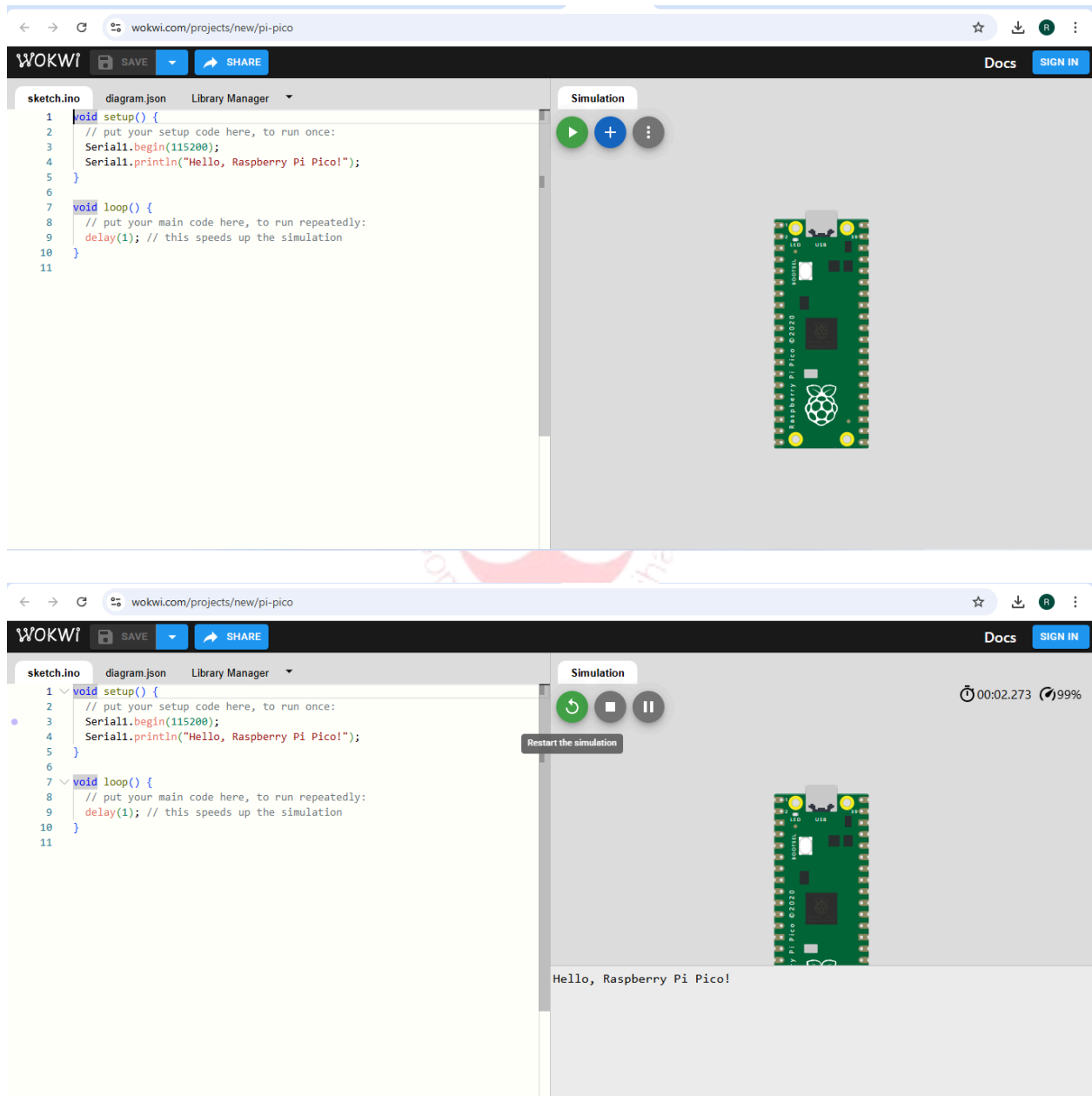
The displacement achieved is commonly linear or rotational, as exemplified by linear motors and rotary motors, respectively. Rotary motion is more natural for small machines making large displacements. By means of a leadscrew, rotary motion can be adapted to function as a linear actuator (a linear motion, but not a linear motor).

Activity:

1. List out sensors which can be use with ESP32.
2. Integrate verity of available sensor in lab with ESP32, list out step performed, circuit, code,

output and components used.

Results: (Program printout with output / Document printout as per the format)



WOKWI

SAVE

SHARE

IoT LAB ThingsSpeak_ESP32_DHT22

by naimish

Docs

sketch.ino

diagram.json

libraries.txt

Library Manager

Simulation

▶

+

⋮

```

1 // channel id : 2411468
2 //channel api key : KVB3DN2C6VSS8VY3
3
4 #include <WiFi.h>
5 #include "DHTesp.h"
6 #include "ThingSpeak.h"
7
8 // Pin configuration
9 const int DHT_PIN = 15; // Pin connected to the DHT sensor
10 const int LED_PIN = 13; // Pin connected to the LED
11
12 // WiFi credentials
13 const char* WIFI_NAME = "Wokwi-GUEST";
14 const char* WIFI_PASSWORD = "";
15
16 // ThingSpeak channel details
17 const int myChannelNumber = 2411468;
18 const char* myApiKey = "KVB3DN2C6VSS8VY3";
19 const char* server = "api.thingspeak.com";
20
21 // DHT sensor and WiFi client
22 DHTesp dhtSensor;
23 WiFiClient client;
24
25 void setup() {
26   Serial.begin(115200);
27
28   // Initialize DHT sensor
29   dhtSensor.setup(DHT_PIN, DHTesp::DHT22);
30
31   // Set up LED pin
32   pinMode(LED_PIN, OUTPUT);
33
34   // Connect to WiFi
35   WiFi.begin(WIFI_NAME, WIFI_PASSWORD);
36   while (WiFi.status() != WL_CONNECTED) {
37     delay(1000);
38     Serial.println("Wifi not connected");
39   }
40   Serial.println("Wifi connected!");
41   Serial.println("Local IP: " + String(WiFi.localIP()));
42
43   // End setup code
44 }

```

thingspeak.mathworks.com/channels/2830135/private_show

Channels Apps Devices Support

Commercial Use How to Buy

Add Visualizations

Add Widgets

Export recent data

MATLAB Analysis

MATLAB Visualization

Channel Stats

Created: 17 minutes ago

Last entry: 6 minutes ago

Entries: 3

Field 1 Chart

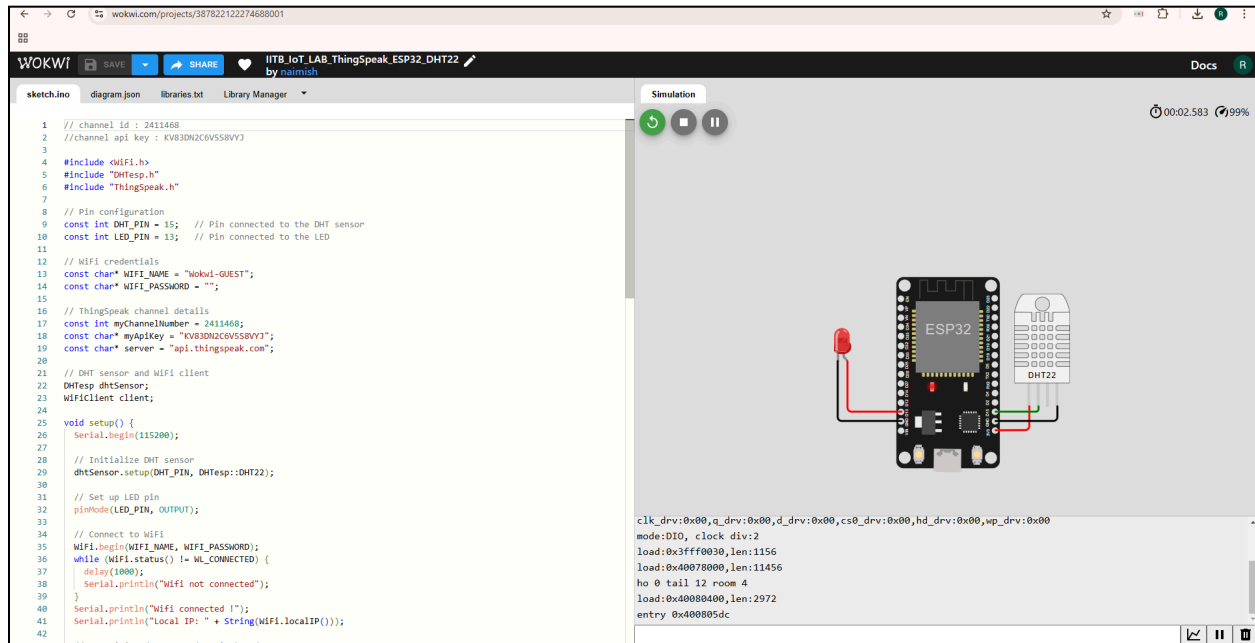
Wifi Connection

Date	Temperature
11:58	25.0
12:05	22.0

Field 2 Chart

Wifi Connection

Date	Humidity
12:05:15	87.0
12:05:45	87.0



Questions:

1. How can ESP32 be used for data analytics?

Ans: The ESP32 chip is a system on a chip (SoC) that is low cost, small, and consumes relatively low power, making it suitable for Internet of Things (IoT) applications. It supports standard IEEE 802.11 security features, including WPA, WPA/WPA2, and WAPI. It also features secure boot and flash encryption. ESP32 can be utilized for data analytics in various ways:

- **Data Processing:** The ESP32 module is capable of processing basic inputs from analog and digital sensors, and can also perform more complex calculations using a real-time operating system (RTOS) or a non-OS software development kit (SDK).
- **Real-time Monitoring** ESP32 can be used for real-time data monitoring in various applications, such as hydroponic farming, photovoltaic systems, and utilities consumption. It captures data from sensors, processes it, and transmits it to a database or cloud platform via Wi-Fi or Bluetooth for remote access and analysis.
- **Data Acquisition** The ESP32 module can acquire analog data via ADC and make it available on a Wi-Fi network through a web server. The web server responds to HTTP queries by returning the value of the data. A microcomputer like Raspberry Pi can then access the ESP32 module to store and manage the data.
- **Industrial IoT:** The ESP32 microcontroller can be integrated into industrial systems for remote monitoring and control, as well as for predictive maintenance.
- **Environmental Monitoring** It can be used to build environmental monitoring systems for measuring air quality, pollution levels, and weather conditions.

ESP32's built-in Wi-Fi and Bluetooth capabilities make it a popular choice for IoT devices, home

automation, wireless communication, robotics, and wearable devices. Its compatibility with Arduino opens the door to numerous libraries, projects, and support from a large community of developers.

Outcomes:

CO-2: Comprehend IoT architecture, enabling technologies and protocols

Conclusion:

Through this experiment, I learnt how to apply the theoretical knowledge of ESP32 into the simulation and saw how the simulation runs on different lines of code. I learned about its compatibility with the Arduino board.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

https://en.wikipedia.org/wiki/Internet_of_things

Books:

1. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014.
2. Vijay Madiseti and Arshdeep Bahga, "Internet of Things (A Hands-on-Approach)", 1st Edition, VPT, 2014.
3. Dr. Ovidiu Vermesan, Dr. Peter Friess, "Internet of Things - From Research and Innovation to Market Deployment", River Publisher, 2014